

Junbin Yuan

Ph.D. Candidate
Department of Mechanical Engineering
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EDUCATION

Carnegie Mellon University Ph.D. in Mechanical Engineering – Advisor: Prof. Sebastian Scherer – Anticipated Graduation: 2027 – QPA: 3.95 / 4.0	09/2022 – Present Pittsburgh, PA, USA
Hong Kong University of Science and Technology B.Eng. in Electronic and Computer Engineering – Minor in Mathematics & Minor in Aerospace Engineering – Cumulative GPA: 3.83 / 4.3, First Class Honors	09/2015 – 07/2019 Hong Kong, China
Massachusetts Institute of Technology Special Student Program – Cumulative GPA: 4.3 / 5.0	01/2018 – 05/2018 Cambridge, MA, USA

APPOINTMENTS

Carnegie Mellon University Research Associate in Robotics Institute – Advisor: Prof. Sebastian Scherer – Contributed to multiple research projects on autonomous aerial robotics, including perception, planning, and system integration.	09/2019 – 03/2022 Pittsburgh, PA, USA
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RESEARCH EXPERIENCE

Semantic-Aware Autonomous Inspection for Construction Sites Ph.D. Student – Leading a research project on autonomous inspection for construction sites, conducted in collaboration with the Institute of Technology, Shimizu Corporation, with a focus on semantic-aware inspection and perception-aware planning in complex environments.	2025 – Present Pittsburgh, PA, USA
Autonomous Mission Execution for Reconnaissance UAVs Research Associate / Ph.D. Student – Supported team research on target search in an Office of Naval Research-funded research project through simulation-based infrastructure development and system integration, enabling large-scale experimental evaluation. – Developed autonomy algorithms for multi-target tracking using UAVs, focusing on decision-making with dynamic targets under uncertainty.	2021 – 2025 Pittsburgh, PA, USA
Autonomous Infrastructure Inspection with Aerial Robots Research Associate – Contributed to a long-term industry-academic collaboration with Institute of Technology, Shimizu Corporation on autonomous infrastructure inspection. Made substantial contributions to the development and maintenance of autonomous flight capabilities, including system integration for sensor calibration, state estimation, and planning modules.	2019 – 2021 Pittsburgh, PA, USA

PUBLICATIONS

Journal Articles

B. Moon, N. Suvarna, A. Jong, S. Chatterjee, **J. Yuan**, M. Cao, and S. Scherer, “IA-TIGRIS: An Incremental and Adaptive Sampling-Based Planner for Online Informative Path Planning,” *IEEE Transactions on Robotics*, 2026.

J. Yuan, B. Moon, M. Cao, and S. Scherer, “Hierarchical Planning for Long-Horizon Multi-Target Tracking Under Target Motion Uncertainty,” *IEEE Robotics and Automation Letters*, 2025.

B. Moon, S. Sachdev, **J. Yuan**, and S. Scherer, “Time-Optimal Path Planning in a Constant Wind for Uncrewed Aerial Vehicles Using Dubins Set Classification,” *IEEE Robotics and Automation Letters*, 2024.

PROFESSIONAL SERVICE

Reviewer

IEEE Transactions on Field Robotics
IEEE Robotics and Automation Letters

TECHNICAL SKILLS

Programming & Systems: C++, Python, MATLAB, Docker
Robotics & Simulation: ROS/ROS2, Gazebo, Isaac Sim
UAV Platforms: PX4, DJI SDK, Ardupilot